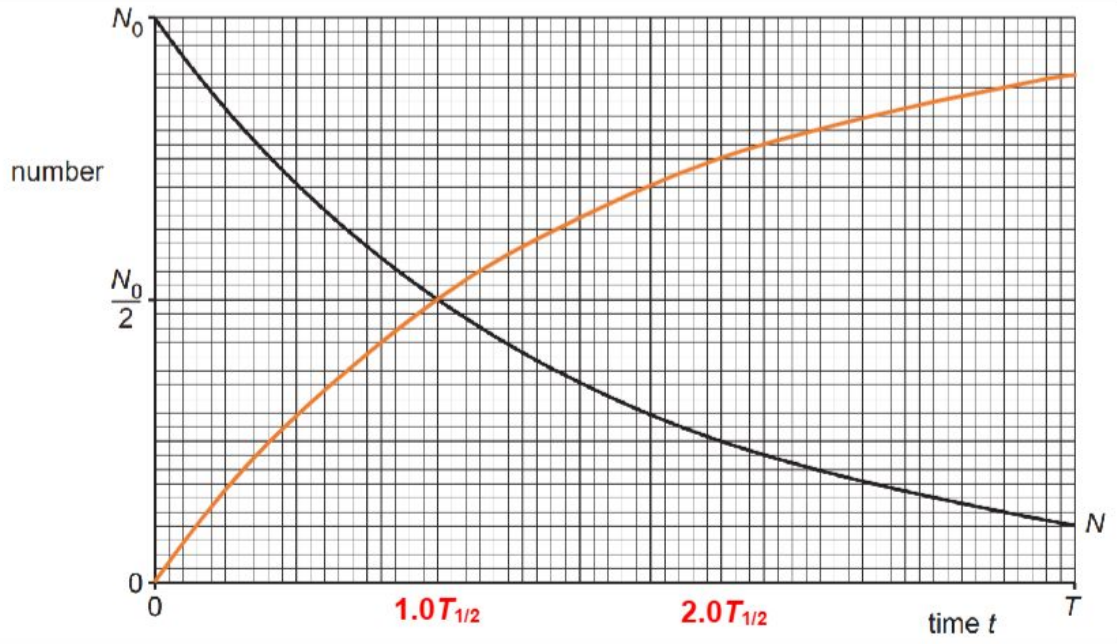
	Hence, the current in coil D flows in the same direction as coil C.	A0
7ai	$\Delta N / \Delta T$	B1
7aii	$\Delta N / N$	B1
7aiii	$\Delta N / (N \Delta T)$	B1
7bi	mass difference = mass of Po – mass of Pb – mass of He = 207.981245u – 203.973043u – 4.002603u = 5.599×10^{-3} u energy = mass difference x c^2 = $(5.599 \times 10^{-3})(1.66 \times 10^{-27})(3.0 \times 10^{8})^2$ = 8.365×10^{-13} J = $(8.365 \times 10^{-13} \times 10^{-6}) / 1.6 \times 10^{-19}$ = 5.228 MeV	A1 C1 A1
7bii	**1.** kinetic energy (of recoil) of the lead (nucleus) **2.** there may be γ radiation	B1 B1
7ci	Mass number = 140 and atomic number = 58	A1
7cii		

	1.0 $T½$ shown at ½ N_0 and 2.0 $T½$ shown at ¼ N_0 Line starting at (0, 0) and reaching points (T, $N_0 - N$), (1.0$T½$, ½ N_0)	B1 B1
7d	Activity of 4 g of fresh volcanic rock = 36 ÷ 3 = 12 Bq Using $A = A_0 e^{-\lambda t}$, $5 = 12e^{-(\frac{\ln 2}{7.8\times10^7})t}$ $\ln(\frac{5}{12}) = -(\frac{\ln 2}{7.8\times10^7})t$ $t = 9.9\times10^7$ years	C1 A1
8a	Gradient = (35.0 – 0.0) / (10.0) = 3.50 m s^{-2}	A1